

Name:

Date:

Period:

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## Projectiles #1

1. A catapult launches a stone into the air at an angle of  $25^\circ$  and an initial resultant velocity of 62 m/s
  - (a) Draw a diagram of the situations and write down knowns and unknowns.
  - (b) Consider that the stone is initially moving at an angle. Find the  $x$ - and  $y$ - components of its initial velocity.
  - (c) How long is the stone in the air?
  - (d) How far in the  $x$ -direction did the stone travel?
  - (e) What is the stone's maximum height?

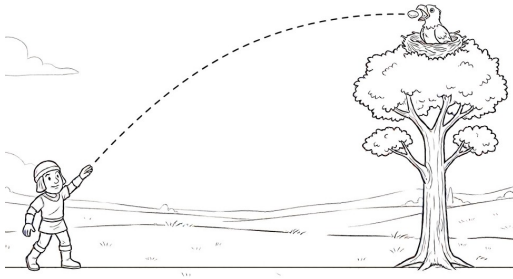
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2. A baby eagle is in a nest atop a 23 m tall tree. You are 13 m away from the base of the tree and need to throw some food up to the bird or it will starve! *Hint: assume that your food will be at its maximum height when it reaches the baby eagle.*

(a) *Label* diagram of the situation, and write down knowns and unknowns.



(b) With what initial  $y$ - velocity must you throw the food?

(c) How much time would it take the food to reach the nest?

(d) What is the initial  $x$ -velocity of the food?

(e) With what initial resultant velocity and at what angle must the food be thrown?